

closure-evidence

BOB-140 — CM-HEALTHCHECK- COVERS-SERVED-PORTS upstreamed into the constitution: closure evidence

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What BOB-140 asked for

`scripts/pre_build/check_cm_healthcheck_covers_served_ports.sh` (landed with BOB-138) implemented a rule every project under this constitution needs — a container health check **MUST** cover every port its service serves, because a check probing a subset asserts a proxy signal instead of the real condition (§11.4.201). Its detection logic already carried zero boba literals, so per §11.4.177 / §11.4.28 / §11.4.74 it belongs in the constitution and must be consumed BY REFERENCE, with boba shipping a thin delegator holding only scope DATA (§11.4.35).

The deferral was deliberate and tracked, not an oversight: a concurrent agent was editing `constitution/scripts/gates/` at the time, and two writers in one submodule directory risks losing work (§11.4.197). That directory was clean at constitution HEAD 997d48f when this work began.

What landed

File	Role
constitution/scripts/gates/ cm_healthcheck_covers_served_ports.sh	NEW — the universal detection ENGINE
constitution/scripts/gates/ cm_healthcheck_covers_served_ports_mutation_test.sh	NEW — the paired \$1.1 mutation test
scripts/pre_build/ check_cm_healthcheck_covers_served_ports.sh	CONVERTED — thin delegator, scope DATA only

config/served_ports.yaml needed no change: the manifest schema the engine consumes is exactly the one boba already declared.

The engine's interface is `--compose <file> --manifest <file> [--quiet]`, with `HEALTHCHECK_PORTS_COMPOSE` / `HEALTHCHECK_PORTS_MANIFEST` / `HEALTHCHECK_PORTS_PYTHON` env equivalents. It carries **no** project literal and refuses (exit 2) rather than guessing a consumer's paths.

The two properties that had to survive the move

Both are load-bearing. Each is now a property of the ENGINE, asserted by the engine's own mutation test rather than by prose:

1. **Zero services checked ⇒ FAIL.** A quiet zero from a blind instrument is indistinguishable from a clean tree (§11.4.201(6)). Two independent routes are covered: an empty manifest, and a manifest whose every serves list is empty (an implementation counting skipped entries as “checked” would pass the second).
2. **No python with PyYAML ⇒ FAIL, never SKIP.** Without a parser the gate cannot assert anything; §11.4.201(4)'s conservative-safe outcome is to refuse and say so honestly.

Neither degraded. Evidence for both is pasted below (sections D and E), taken through the delegator on boba's real tree.

A. Output equivalence with the pre-move implementation

Both spellings run against the same real compose file and manifest. The old implementation is `git show HEAD:` of the pre-conversion script, run with explicit paths (a copy outside the repo cannot derive boba's `PROJECT_ROOT`).

```
old rc=0
new rc=0
--- stdout diff (empty == byte-identical) ---
STDOUT IDENTICAL
--- stderr diff ---
STDERR IDENTICAL
--- md5 ---
bd38e61fe453938dd28fb971c927c5cb /tmp/out_old.txt
bd38e61fe453938dd28fb971c927c5cb /tmp/out_new.txt
```

The invocation contract is therefore preserved exactly: same path, same optional positional [`COMPOSE_FILE`] [`MANIFEST`], same exit codes, same `PASS`: / `FAIL`: line shapes, and the `PASS` line is still **last** — which matters, because pre-build invariant 44 reads it with `tail -n1`.

B. Delegator on boba's real tree

```
$ bash scripts/pre_build/
check_cm_healthcheck_covers_served_ports.sh
  ok qbittorrent: healthcheck covers all served ports [7185]
  ok jaxxnet: healthcheck covers all served ports [9117]
  ok download-proxy: healthcheck covers all served ports [7186,
7187]
  ok qbittorrent-proxy-go: healthcheck covers all served ports
[7187]
  ok boba-jaxxnet: healthcheck covers all served ports [7189]
CM-HEALTHCHECK-COVERS-SERVED-PORTS: PASS (5 services verified)
rc=0
```

C. Paired §1.1 mutation test — both directions

13 fixtures, 11 FAIL-on-mutation and 2 PASS-on-clean negative controls.

```
=====
§1.1 paired-mutation meta-test for CM-HEALTHCHECK-COVERS-SERVED-
PORTS
anchors: §11.4.201 (guard asserts the real condition) / §11.4.254
fixtures under: /tmp/.private/milosvasic/hcports_mut.thRv7J
=====
    META OK:   BOB-138 literal shape (serves 7186+7187, probes only
7186) – gate correctly FAILED on the mutation
    META OK:   BOB-138 fixed shape (health check probes BOTH served
ports) – gate correctly PASSEd on clean fixture
    META OK:   declared service with NO healthcheck at all – gate
correctly FAILED on the mutation
    META OK:   substring near-miss (serves 7187, probes 71870 –
must NOT match inside) – gate correctly FAILED on the mutation
    META OK:   UNDECLARED compose service with a healthcheck +
ports – gate correctly FAILED on the mutation
    META OK:   STALE manifest entry (declared but absent from the
compose file) – gate correctly FAILED on the mutation
    META OK:   ZERO services checked (empty manifest) – blind, not
clean – gate correctly FAILED on the mutation
    META OK:   ZERO services checked (all 'serves' lists empty) –
blind, not clean – gate correctly FAILED on the mutation
    META OK:   no python with PyYAML – BLIND gate must FAIL, not
skip – gate correctly FAILED on the mutation
    META OK:   missing compose file (blind input) – gate correctly
FAILED on the mutation
    META OK:   missing manifest file (blind input) – gate correctly
FAILED on the mutation
    META OK:   negative control: every service fully covered (list
+ string test forms) – gate correctly PASSEd on clean fixture
    META OK:   negative control: port-less worker absent from
manifest is not 'undeclared' – gate correctly PASSEd on clean
fixture
=====
    META PASS – CM-HEALTHCHECK-COVERS-SERVED-PORTS FAILs-on-
mutation AND PASSes-on-clean for every fixture (§1.1 proof holds)
META rc=0
```

The two negative controls exist because §11.4.201(1) makes a false-positive refusal a FAIL-bluff exactly as a false pass is a PASS-bluff. Control 1 is a multi-service fixture where every health check legitimately covers all its served ports, exercising both test: spellings (list form and bare string form) plus a genuine multi-port probe.

Control 2 pins the undeclared-service rule's boundary: a port-less worker absent from the manifest publishes nothing and must not be reported.

D. Zero-services case, through the delegator

```
$ bash scripts/pre_build/
check_cm_healthcheck_covers_served_ports.sh "$T/compose.yml" "$T/
manifest.yml"
CM-HEALTHCHECK-COVERS-SERVED-PORTS: FAIL – checked 0 services; the
gate is blind (empty manifest or no matching services), not clean
rc=1 <-- MUST be 1 (blind, not clean)
```

E. Missing-PyYAML case, through the delegator

The delegator's interpreter candidates include an **absolute** `.venv/bin/python` path, so `PATH` shadowing alone cannot reach it. The fixture therefore builds a throw-away project root containing the real delegator, the real constitution (by symlink) and byte-identical copies of the real compose file and manifest — but no `.venv`, so every candidate is `PATH`-reachable and shadowable. The sanity run proves the fixture `PASSes` with a real interpreter first, so the subsequent `FAIL` is attributable to the interpreter and nothing else.

```
Sanity: this throw-away root PASSes with a real interpreter –
CM-HEALTHCHECK-COVERS-SERVED-PORTS: PASS (5 services verified)
rc=0
```

Now shadow `python3/python` with interpreter-shaped executables that have no PyYAML.

ONLY the interpreter changes – the compose file and manifest are byte-identical.

```
CM-HEALTHCHECK-COVERS-SERVED-PORTS: FAIL – no python interpreter
with PyYAML available
```

```
  tried: /tmp/.private/milosvasic/tmp.JMcPyCC3d7/root/.venv/bin/
python python3 python
```

the gate cannot parse the compose file, so it cannot assert anything.

This is a `BLIND` gate, not a `clean tree` (§11.4.201(6)).

```
rc=1 <-- MUST be 1: a BLIND gate FAILs, it never skips and
never passes
```

F. Real mutation on boba's own compose file, through the delegator

Reverting the download-proxy health check to its exact pre-BOB-138 state — one line changed, nothing else — must be refused.

```
--- the ONLY difference from the real compose file ---
230c230
<      test: ["CMD-SHELL", "python -c \"import urllib.request as
u; u.urlopen('http://localhost:7186/', timeout=5);
u.urlopen('http://localhost:7187/health', timeout=5)\" || exit 1"]
---
>      test: ["CMD-SHELL", "python -c \"import urllib.request as
u; u.urlopen('http://localhost:7186/', timeout=5)\" || exit 1"]

--- delegator against the MUTATED compose (real manifest,
unchanged) ---
CM-HEALTHCHECK-COVERS-SERVED-PORTS: FAIL
- download-proxy: serves [7186, 7187] but its healthcheck probes
none of [7187]
    healthcheck: CMD-SHELL python -c "import urllib.request as
u; u.urlopen('http://localhost:7186/', timeout=5)" || exit 1
    -> a dead port in [7187] reports HEALTHY forever (§11.4.201
proxy signal)
    ok qbittorrent: healthcheck covers all served ports [7185]
    ok jackett: healthcheck covers all served ports [9117]
    ok qbittorrent-proxy-go: healthcheck covers all served ports
[7187]
    ok boba-jackett: healthcheck covers all served ports [7189]
    rc=1 <-- MUST be 1

--- G. paired control: same command, UNMUTATED compose ---
    ok qbittorrent: healthcheck covers all served ports [7185]
    ok jackett: healthcheck covers all served ports [9117]
    ok download-proxy: healthcheck covers all served ports [7186,
7187]
    ok qbittorrent-proxy-go: healthcheck covers all served ports
[7187]
    ok boba-jackett: healthcheck covers all served ports [7189]
CM-HEALTHCHECK-COVERS-SERVED-PORTS: PASS (5 services verified)
    rc=0 <-- MUST be 0
```

H. No duplicated detection logic (§11.4.177)

The delegator's complete executable body — 19 lines, comments and blanks stripped — is engine resolution, scope DATA and one exec:

```

1  set -euo pipefail
2  SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
3  PROJECT_ROOT="$(cd "${SCRIPT_DIR}/../.." && pwd)"
4  CONST_GATE="${PROJECT_ROOT}/constitution/scripts/gates/
cm_healthcheck_covers_served_ports.sh"
5  if [[ "${1:-}" == "--help" || "${1:-}" == "-h" ]]; then
6      sed -n '2,80p' "${BASH_SOURCE[0]}" | sed 's/^#
\{0,1\}\\/'
7      exit 0
8  fi
9  if [[ ! -f "$CONST_GATE" ]]; then
10     echo "CM-HEALTHCHECK-COVERS-SERVED-PORTS: ERROR –
shared gate engine missing: $CONST_GATE" >&2
11     echo " ($11.4.177 – this gate is consumed by
reference; run" >&2
12     echo " 'git submodule update --init constitution' to
restore it)" >&2
13     exit 2
14  fi
15  COMPOSE_FILE="${1:-${PROJECT_ROOT}/docker-compose.yml}"
16  MANIFEST="${2:-${PROJECT_ROOT}/config/served_ports.yaml}"
17  PY_CANDIDATES="${PYTHON_BIN:-} ${PROJECT_ROOT}/.venv/bin/
python python3 python"
18  exec env HEALTHCHECK_PORTS_PYTHON="$PY_CANDIDATES" \
19      bash "$CONST_GATE" --compose "$COMPOSE_FILE" --
manifest "$MANIFEST"

```

Detection-primitive census over executable lines only:

primitive	delegator	engine
yaml	1	5
safe_load	0	2
re.search	0	1
healthcheck	1	7
serves	0	6
PYEOF	0	2
UNDECLARED	0	1
findings	0	7
checked	0	5

The delegator's two non-zero counts are **file paths, not logic**: config/served_ports.yaml on line 16 and the engine filename on lines 4 and 10. Every detection primitive lives in exactly one place.

I. Invariant 44 wiring intact

scripts/pre_build_verification.sh was **not** edited. Its exact code path, replayed:

```
CONST_GATE_TIMEOUT = 300s
    PASS    CM-HEALTHCHECK-COVERS-SERVED-PORTS: CM-HEALTHCHECK-
COVERS-SERVED-PORTS: PASS (5 services verified)
            (747ms, well inside the 300s budget)
```

The doubled gate name is pre-existing formatting in invariant 44's pass call, not a regression — it prefixes the label onto `tail -n1` of the gate's own output, and the gate's output has always begun with its own name.

Syntax checks

```
engine bash -n OK
mutation-test bash -n OK
delegator bash -n OK
```

Honest boundary (§11.4.6)

Stated as facts, not assumed:

- **PYTHON_BIN semantics changed, deliberately and for the stronger.** The old script trusted a set PYTHON_BIN on sight, with no `import yaml` probe. It is now the first candidate but is probed like any other, so a PYTHON_BIN lacking PyYAML falls through to the next candidate instead of failing confusingly downstream (§11.4.201(11): probe the artifact through its real invocation path). This is documented in the delegator header, not silent.
- **A new exit code 2 exists** for a missing engine or an engine usage error. Invariant 44 treats any non-zero, non-124 status as FAIL, so a missing constitution submodule blocks the build rather than passing — the correct direction. Exit codes 0 and 1 are unchanged.
- **Docker-compose parsing is unchanged**, including the shape it does not handle: the engine reads `services.<name>.healthcheck.test` and `services.<name>.ports` from a single compose file. It does not resolve extends, multi-file overlays, or profiles. That limit existed before this move and is not newly introduced; it is not currently exercised by boba.

- **The mutation test needs PyYAML to run.** If absent it exits 2 (environment error), never 0 — an unseeing meta-test must not report success.
- **docs/scripts/check_cm_healthcheck_covers_served_ports.md does not exist.** §11.4.18 requires a companion user guide for every shell script. This gap pre-dates BOB-140 (the pre-move script's header cross-referenced the document as though it existed). The delegator's header now states the gap honestly instead of citing a non-existent file; boba has no CM-SCRIPT-DOCS-SYNC invariant wired in scripts/pre_build_verification.sh, so nothing mechanical catches it. **This remains open and is not claimed closed** — docs/scripts/ is outside this work item's file ownership, so it was not created here.
- **Nothing was committed or pushed** in either repository, and no container was restarted, per this work item's scope.

Verdict

CM-HEALTHCHECK-COVERS-SERVED-PORTS now lives in the constitution as a project-agnostic engine with a paired §1.1 mutation test proving both polarities; boba consumes it by reference through a 19-line delegator carrying only scope DATA; the invocation contract and output are byte-identical to the pre-move implementation; and both load-bearing FAIL properties — zero services checked, and no usable YAML parser — survive the move, proven end-to-end through the delegator.